

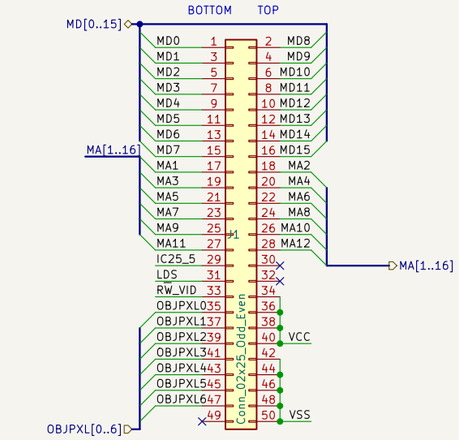


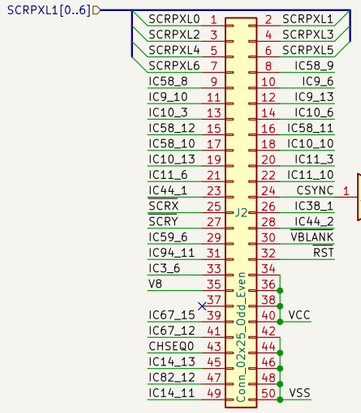
Leo Tejada
JOTEGO

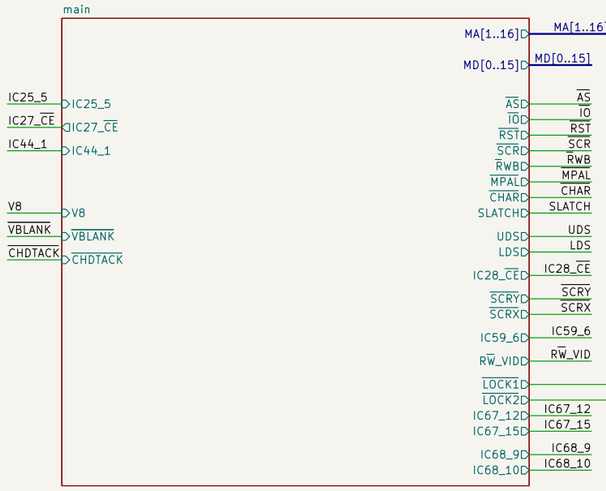
Sheet: /
File: wwfsstar.kicad_sch

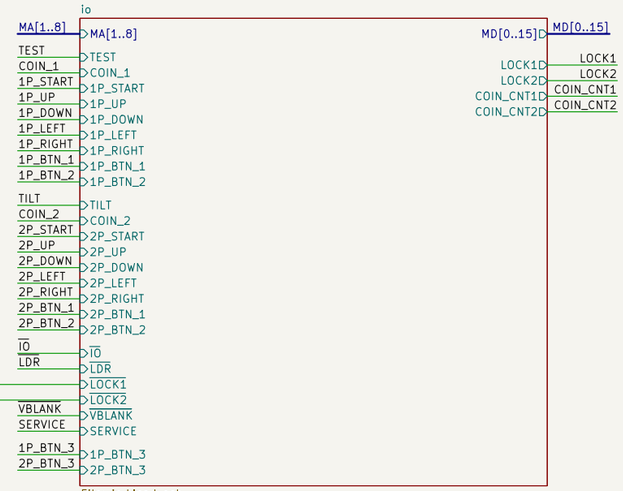
Title: WWF Super Stars

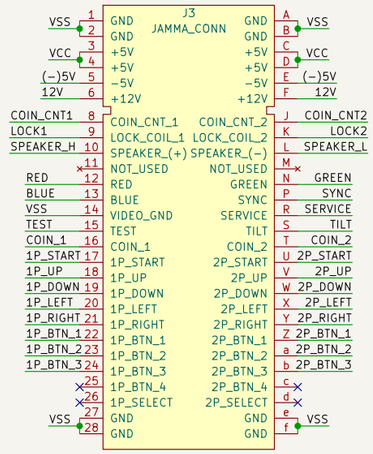
Size: A3	Date:	**Rev:**
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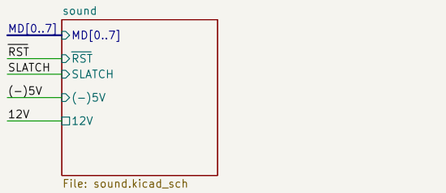


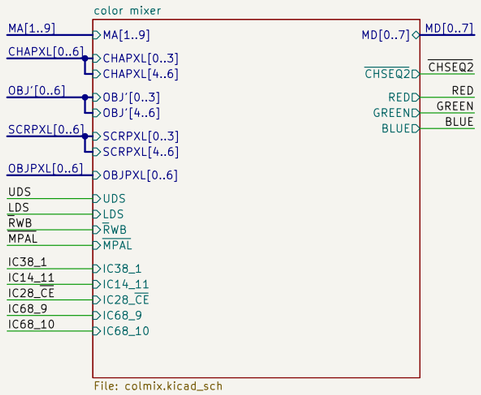


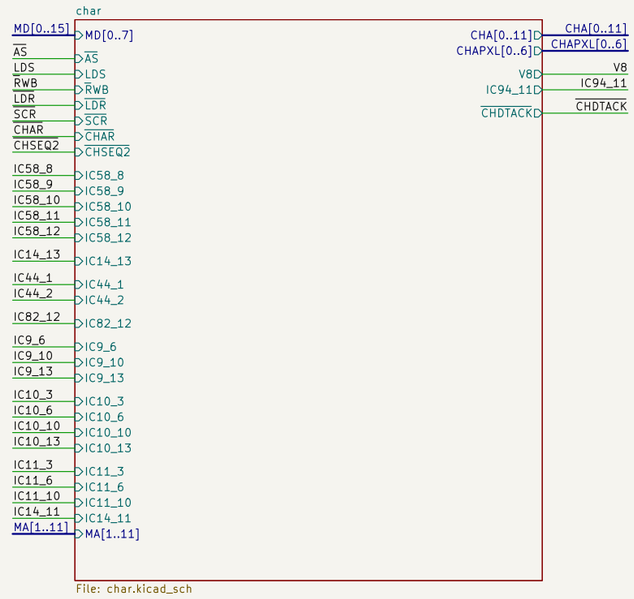












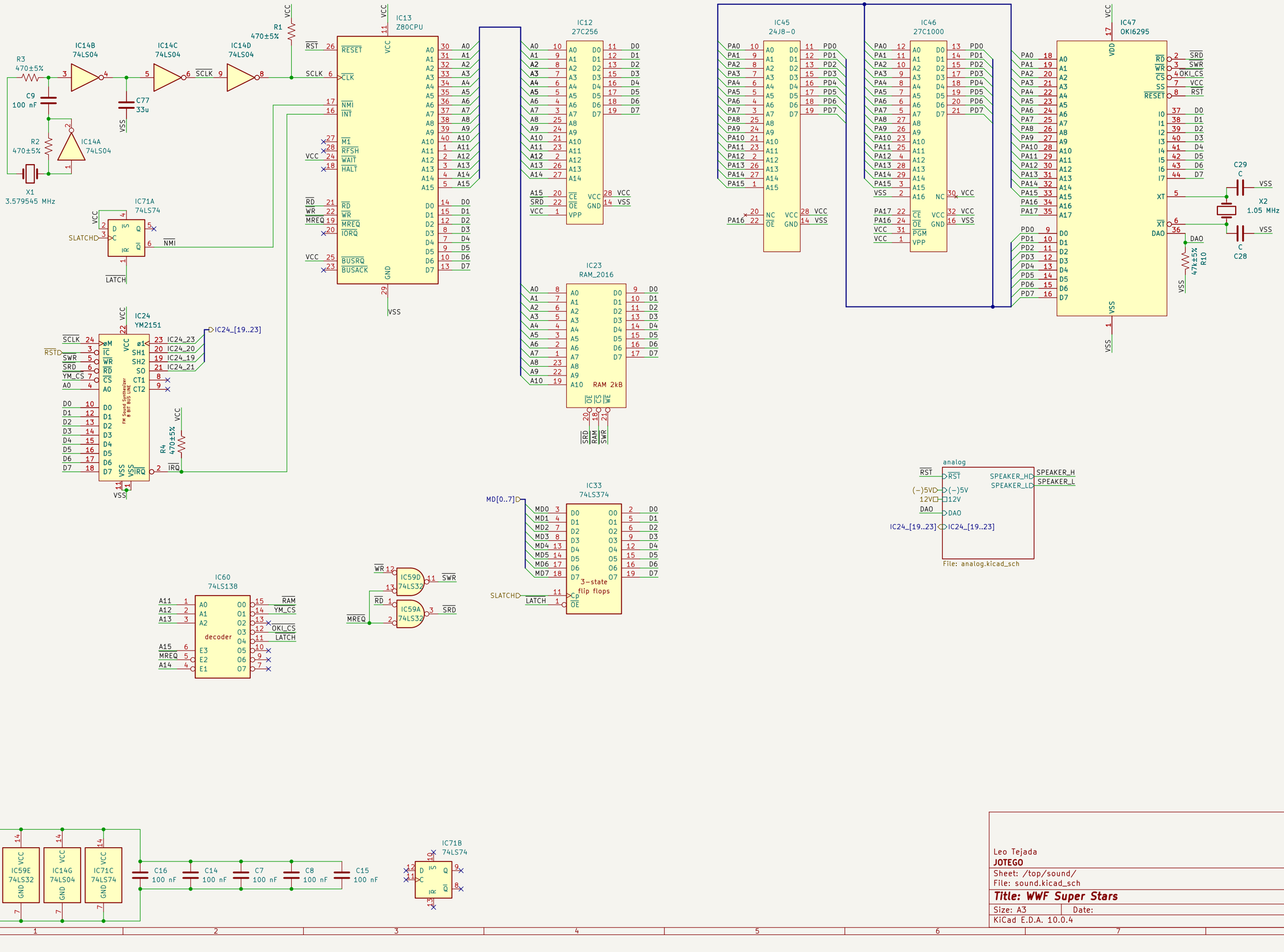
JOTEGO

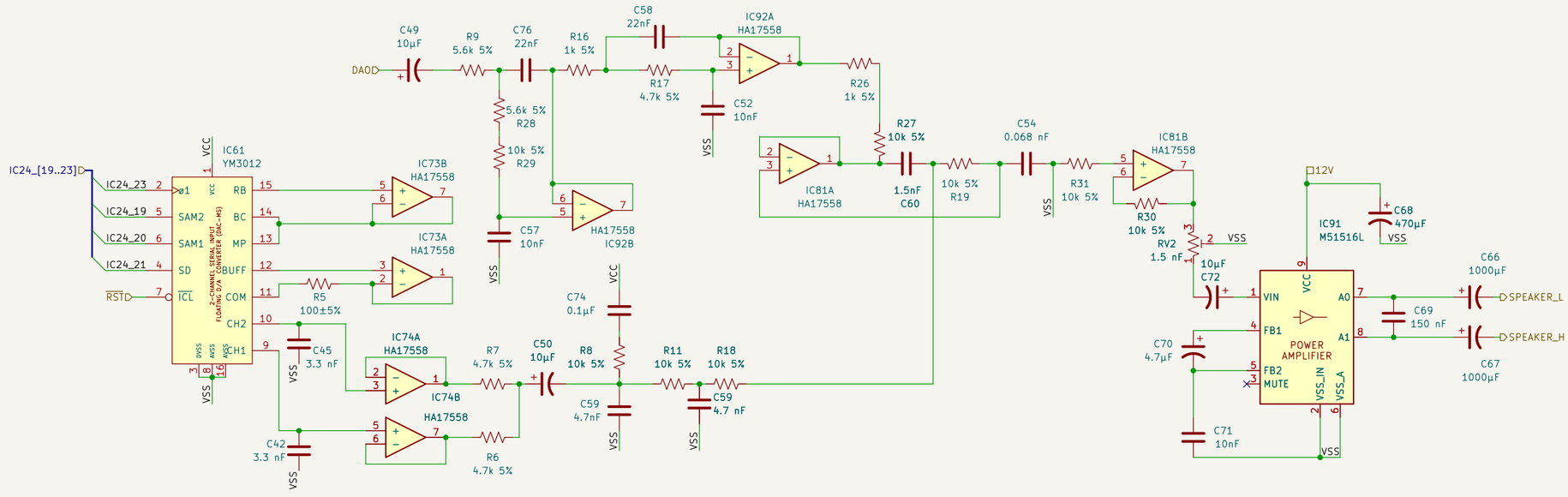
Sheet: /top/

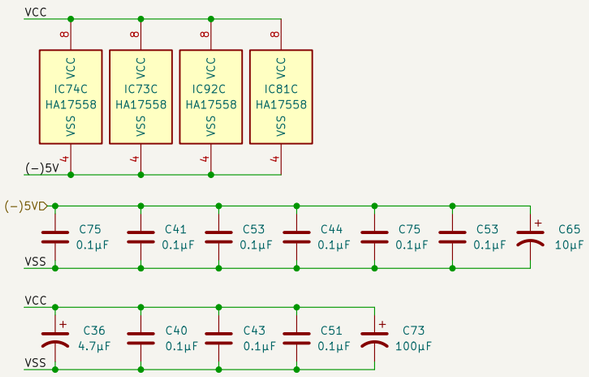
File: top.kicad_sch

Title: Connectors and subsystems

Size: A3	Date:	Rev:
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IC13
Z80CPU
IC12
27C256
IC45
2408-0
IC46
27C1000
IC47
OKI6295
IC14B
74LS04
IC14C
74LS04
IC14D
74LS04
IC14A
74LS04
R1
470±5%
R2
470±5%
R3
470±5%
C9
100 nF
C77
33u
X1
3.579545 MHz
IC71A
74LS74
SLATCHD
LATCH
IC24
YM2151
FM Sound Synthesizer 8 Bit BUS Type
IC24_[19..23]
R4
470±5%
IC23
RAM_2016
RAM 2kB
IC33
74LS374
MD[0..7]
3-state flip flops
SLATCH
IC60
74LS138
decoder
IC59D
74LS32
IC59A
74LS32
C29
C
X2
1.05 MHz
C28
R10
47k±5%
analog
SPEAKER_H
SPEAKER_L
(-)5V
12V
DAO
IC24_[19..23]
File: analog.kicad_sch
IC84G
74LS14
IC59E
74LS32
IC14G
74LS04
IC71C
74LS74
C16
100 nF
C14
100 nF
C7
100 nF
C8
100 nF
C15
100 nF
IC71B
74LS74
Leo Tejada
JOTEGO
Sheet: /top/sound/
File: sound.kicad_sch
Title: WWF Super Stars
Size: A3
Date:
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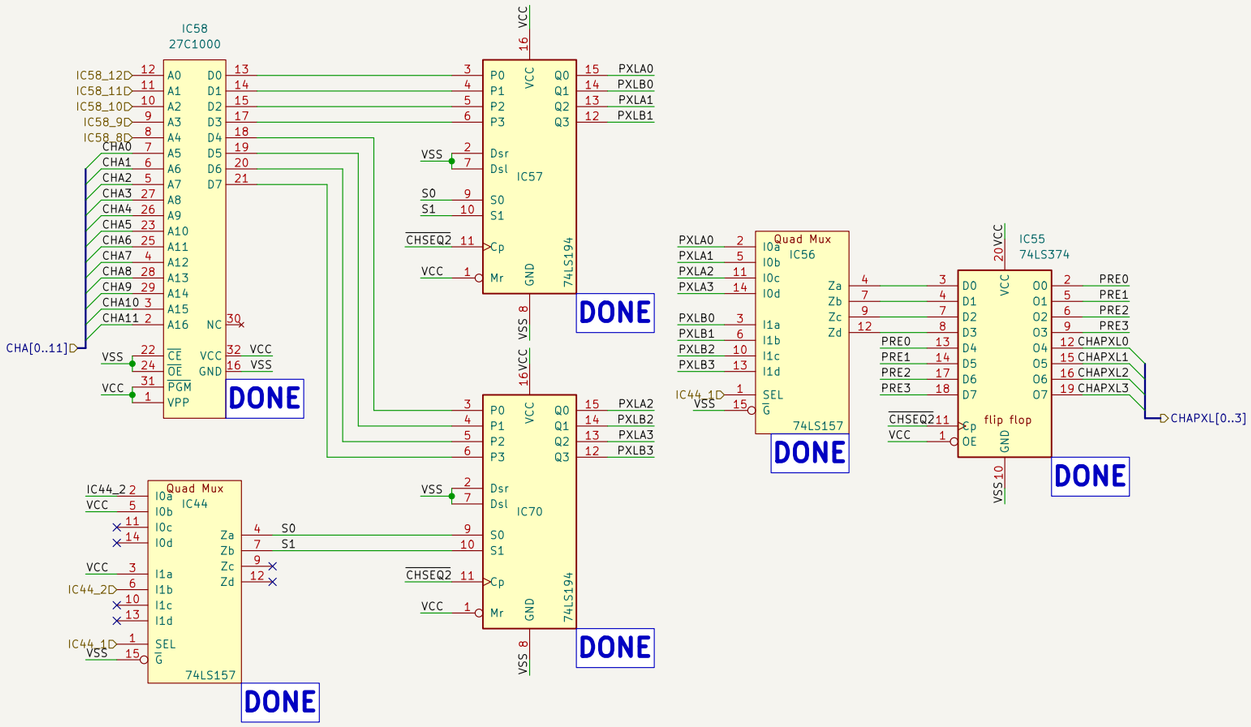
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File: analog.kicad_sch

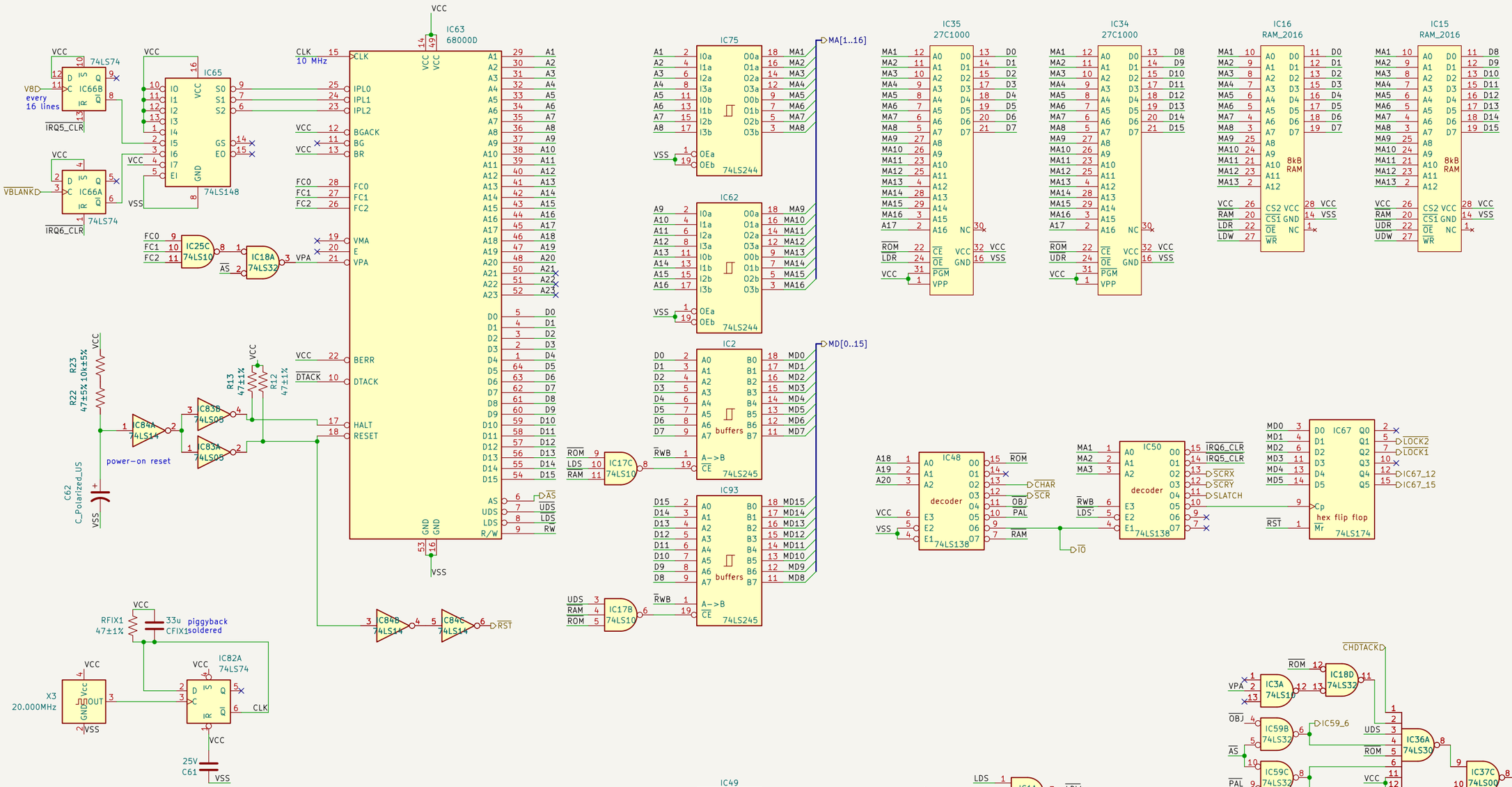
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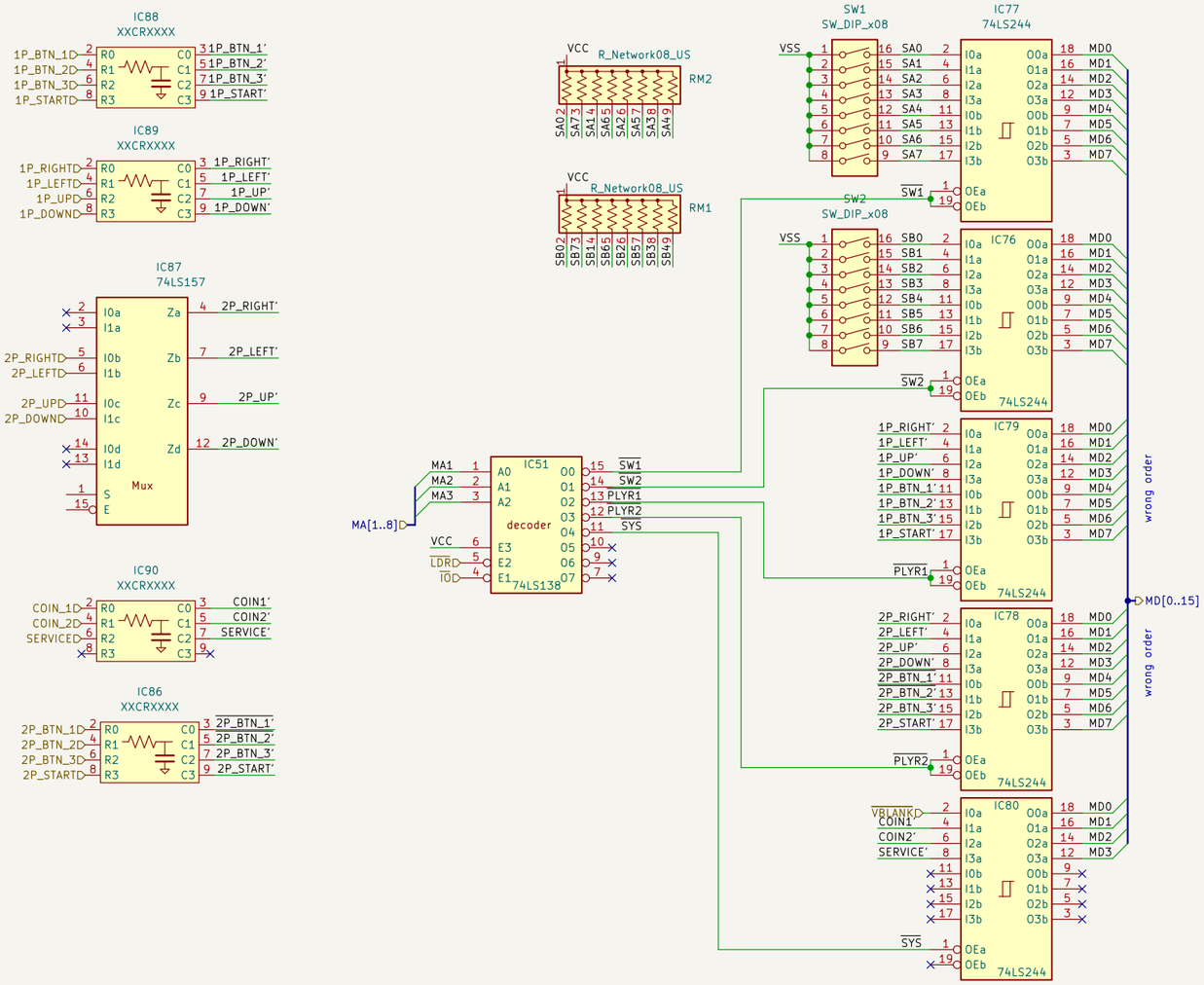
Size: A3	Date:	**Rev:**
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MA[1..11]
MD[0..7]
CHA[0..11]
RWB
MCHAR
CHSEQ0
CHSEQ1
CHSEQ2
LDR
SEL
VSS
VCC
RAMOE
RAMWE
SCR
CHAR
LDS
AS
V8
PRECHDTACK
PREPXL[0..3]
CHAPXL[0..6]
IC14_13
IC82_12
IC14_11
IC94_11
IC11_3D
IC11_6D
IC11_10D
IC9_6D
IC9_10D
IC9_13D
IC10_3D
IC10_6D
IC10_10D
IC10_13D
MA1
MA2
MA3
MA4
MA5
MA6
MA7
MA8
MA9
MA10
MA11
A0
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
MD0
MD1
MD2
MD3
MD4
MD5
MD6
MD7
D0
D1
D2
D3
D4
D5
D6
D7
CHA0
CHA1
CHA2
CHA3
CHA4
CHA5
CHA6
CHA7
CHA8
CHA9
CHA10
CHA11
PREPXL0
PREPXL1
PREPXL2
PREPXL3
CHAPXL0
CHAPXL1
CHAPXL2
CHAPXL3
CHAPXL4
CHAPXL5
CHAPXL6
CHDTACK
IC44_1
IC44_2
IC58_8
IC58_9
IC58_10
IC58_11
IC58_12
Quad Mux
74LS157
IC9
IC10
IC11
IC32
buffers
A->B
74LS245
IC22
RAM_2016
RAM 2kB
IC42
IC43
IC41
IC40
flip flops
hex flip flop
74LS273
74LS174
IC14F
74LS04
IC3F
74LS10
IC26A
IC26B
74LS74
IC82B
IC68A
74LS00
IC18C
74LS32
IC37B
IC94D
IC3C
IC69A
IC68D
char pixels
File: chpixel.kicad_sch
IC69D
IC68E
IC37E
IC18E
IC94E
IC26C
IC36B
IC25D
IC3G
IC72G
IC82C
IC3D
GND
JOTEGO
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Title:
Size: A3
Date:
Rev:
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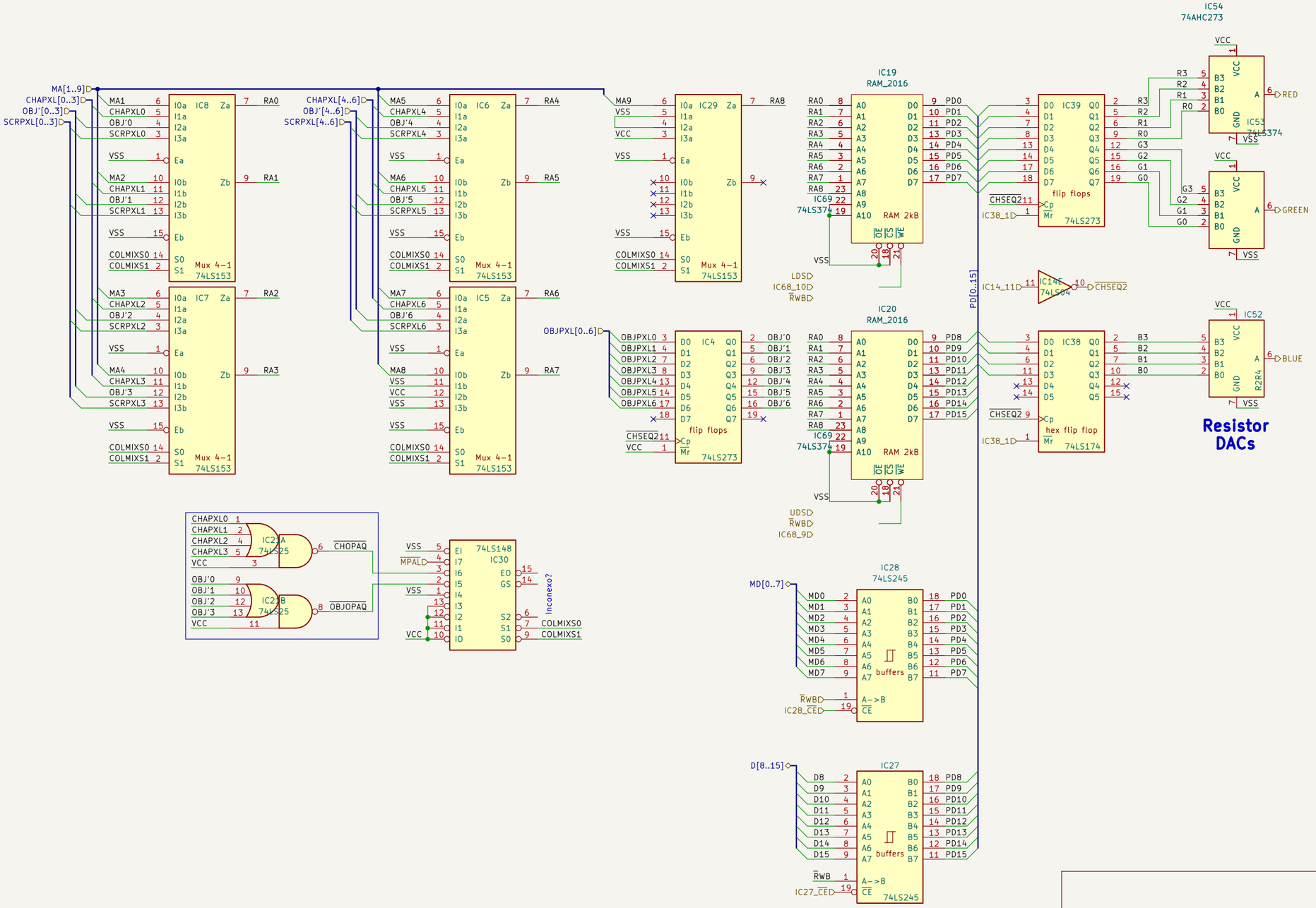


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Title:		
Size: A3	Date:	Rev:
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IC63
68000D
CLK
10 MHz
IPL0
IPL1
IPL2
BGACK
BG
BR
FC0
FC1
FC2
VMA
E
VPA
BERR
DTACK
HALT
RESET
AS
UDS
LDS
R/W
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
A11
A12
A13
A14
A15
A16
A17
A18
A19
A20
A21
A22
A23
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
D12
D13
D14
D15
IC65
74LS148
IC66B
IC66A
74LS74
IRQ5_CLR
IRQ6_CLR
V8D
every
16 lines
VBLANK
IC25C
74LS10
IC18A
74LS32
R22
47±5%
10k±5%
R23
R13
47±1%
R12
47±1%
IC84A
74LS14
IC83B
74LS05
IC83A
power-on reset
C62
C_Polarized_US
IC75
IC62
IC2
IC93
IC49
74LS244
74LS245
buffers
A->B
MA[1..16]
MD[0..15]
IC17C
74LS10
IC17B
IC35
27C1000
IC34
IC16
RAM_2016
IC15
8kB
RAM
IC48
decoder
74LS138
IC50
IC67
hex flip flop
74LS174
LOCK2
LOCK1
IC67_12
IC67_15
CHAR
SCR
OBJ
PAL
SCRX
SCRY
SLATCH
IRQ6_CLR
IRQ5_CLR
RAM
ROM
IO
RFIX1
47±1%
33u
CFIX1
piggyback
soldered
X3
20.000MHz
IC82A
74LS74
CLK
25V
C61
IC64F
IC83C
IC72D
IC72F
IC17A
IC1B
IC1C
IC18B
IC64G
IC1E
IC17D
IC66C
IC83G
IC21C
74LS04
74LS00
74LS10
74LS32
74LS05
74LS25
IC64H
IC64G
IC64E
IC1A
IC37D
IC37A
IC25_5
RWB
RW_VID
LDW
UDW
IC44_1D
CHDTACK
IC3A
IC18D
IC59B
IC59C
IC36A
IC37C
IC25B
IC1D
IC72B
IC68C
IC68B
IC68_10
IC68_9
IC28_CE
IC27_CE
IC59_6
MPAL
JOTEGO
Sheet: /top/main/
File: main.kicad_sch
Title:
Size: A3
Date:
Rev:
KiCad E.D.A. 10.0.4
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JOTEGO	
Sheet: /top/io/	
File: io.kicad_sch	
Title: CABINET I/O	
Size: A3	Date:
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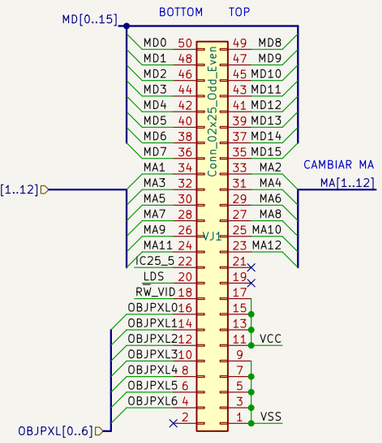


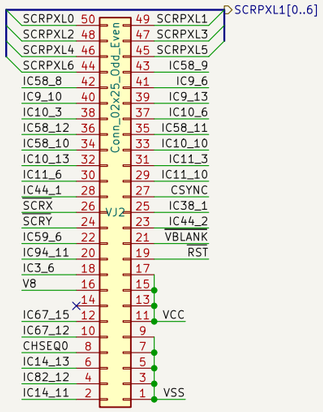
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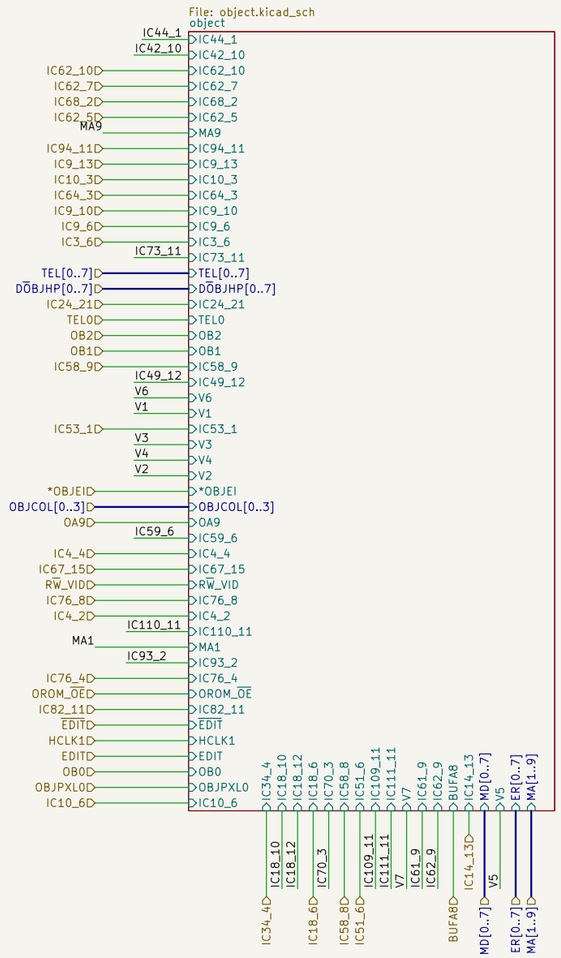
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File: colmix.kicad_sch

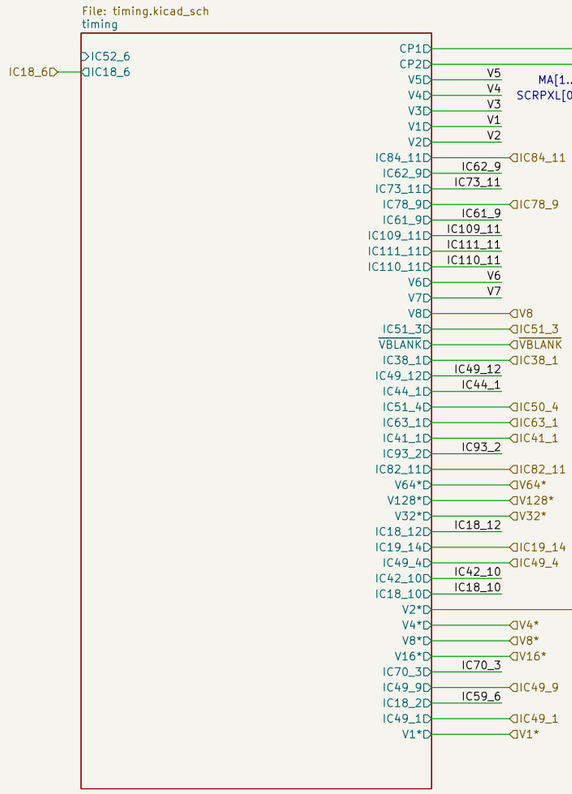
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Size: A3	Date:	**Rev:**
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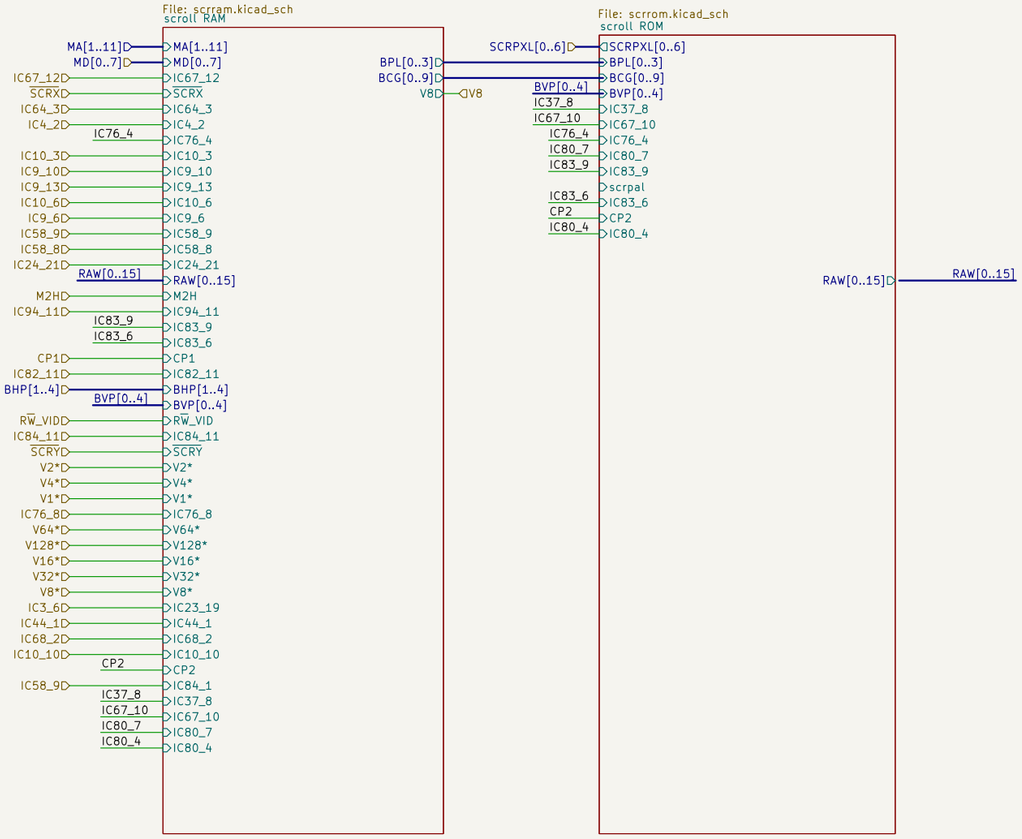








JOTEGO		
Sheet: /bottom/		
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Title: Bottom board		
Size: A3	Date:	**Rev:**
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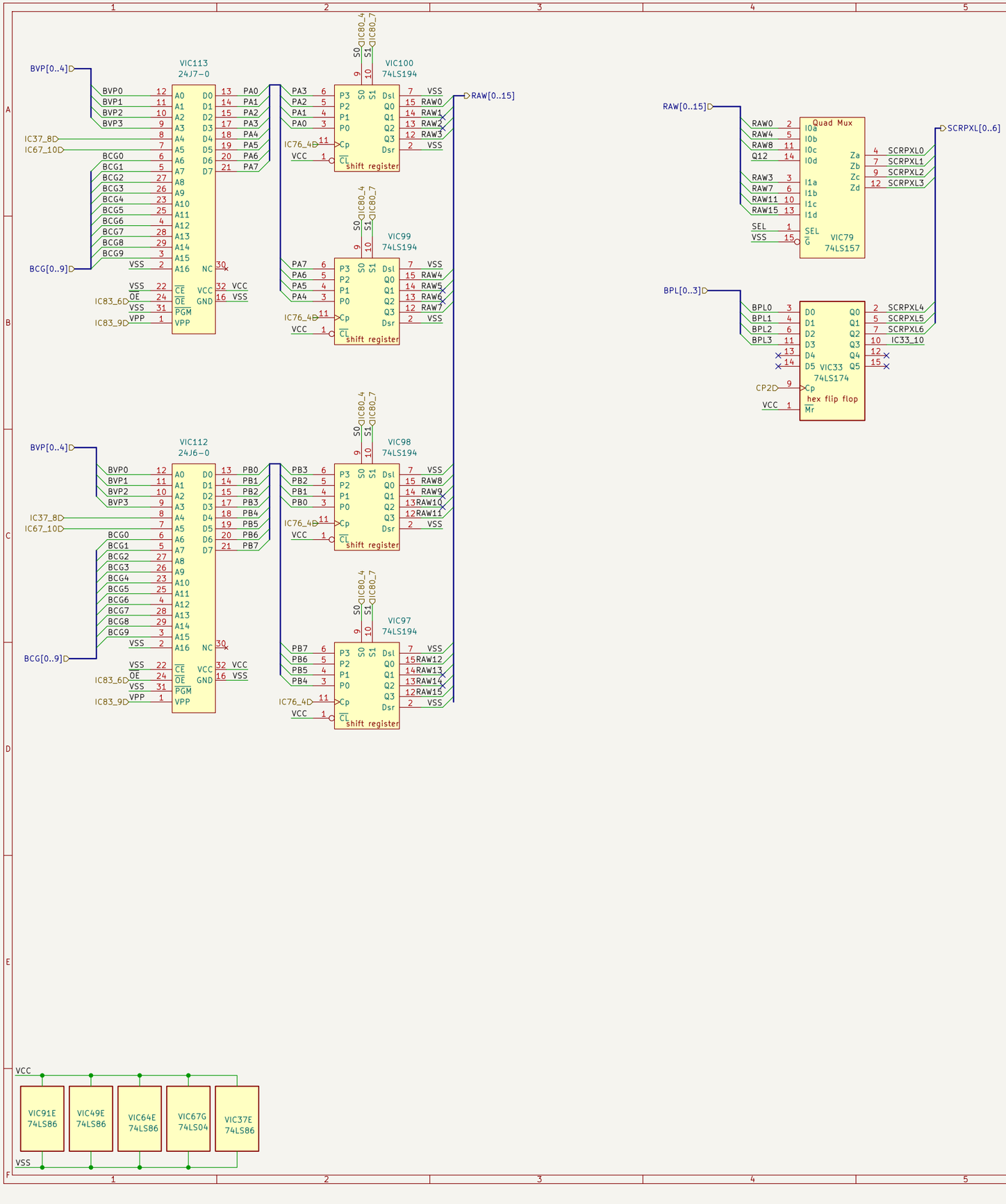


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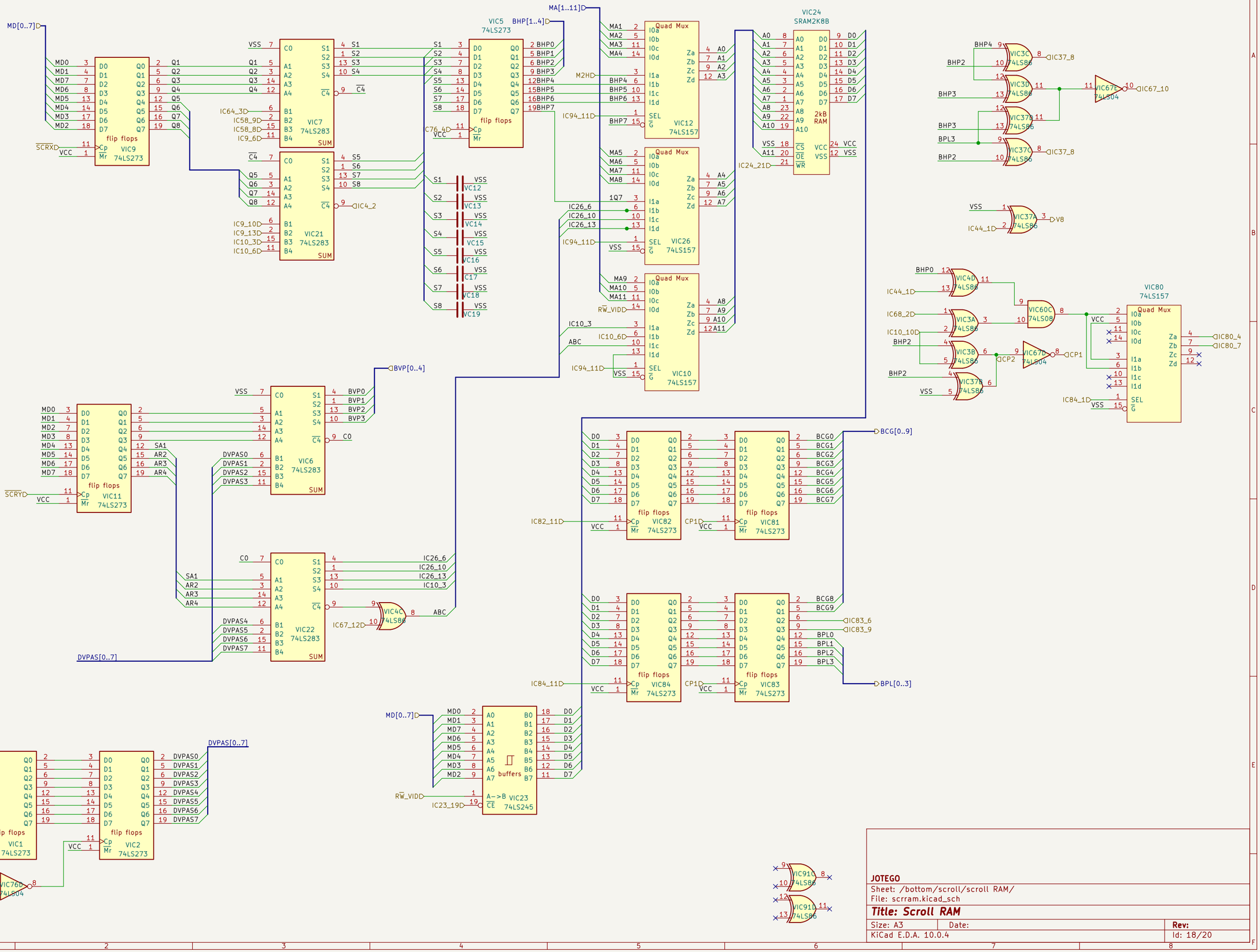
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File: scroll.kicad_sch

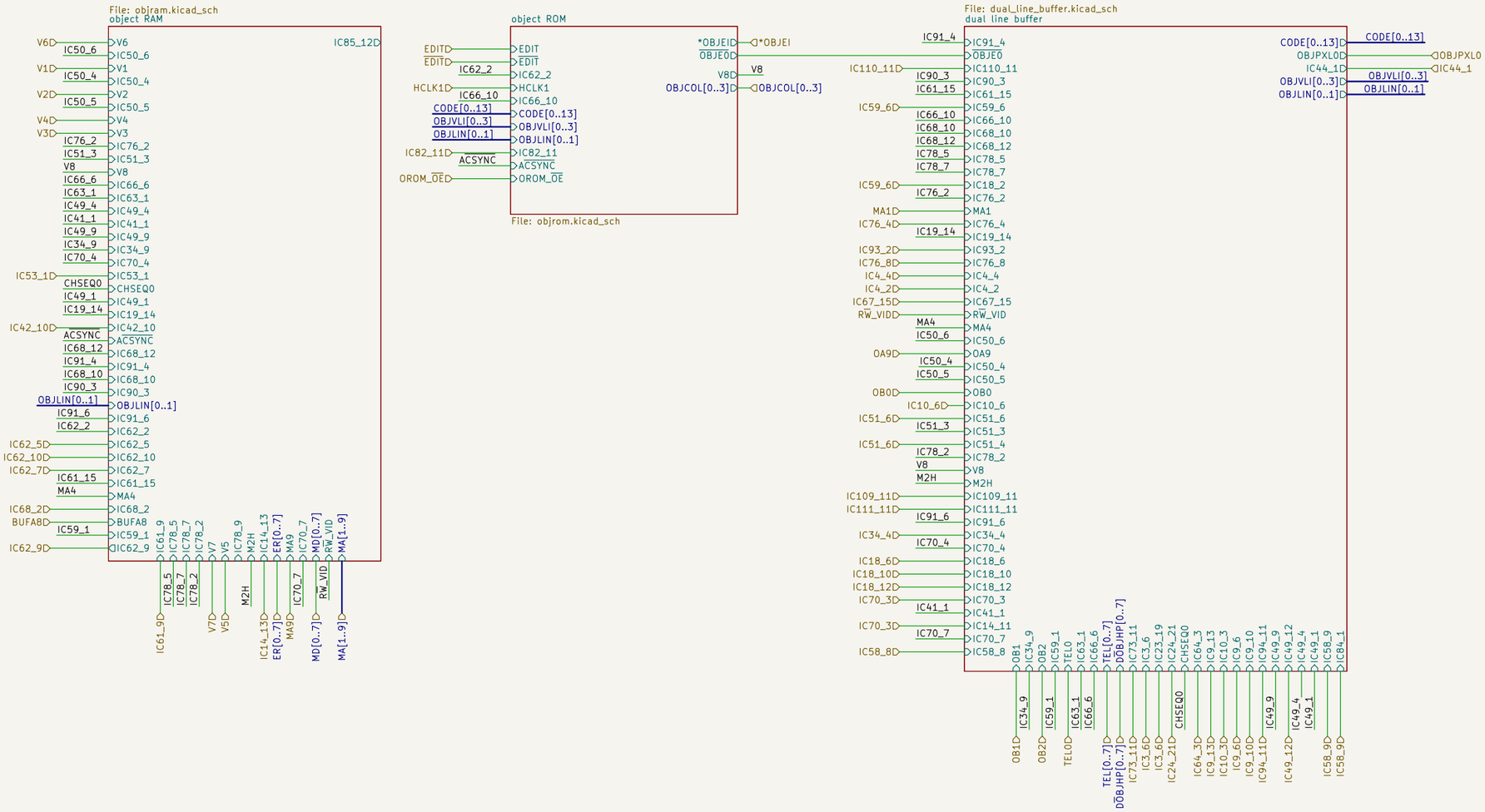
Title:

Size: A3	Date:	**Rev:**
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JOTEGO	
Sheet: /bottom/scroll/scroll ROM/	
File: scrrom.kicad_sch	
Title:	
Size: A3	Date:
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MD[0..7]
SCRXD
VIC9 74LS273
flip flops
VIC7 74LS283
VIC21 74LS283
SUM
IC4_2
MA[1..11]
VIC5 74LS273
BHP[1..4]
VIC12 74LS157
VIC26 74LS157
VIC10 74LS157
Quad Mux
VIC24 SRAM2K8B
2kB RAM
M2HD
IC94_11
IC24_21
IC26_6
IC26_10
IC26_13
IC10_3
IC10_6
ABC
RW_VID
VIC4C 74LS86
IC67_12
BVP[0..4]
VIC6 74LS283
VIC11 74LS273
VIC22 74LS283
DVPAS[0..7]
SCRY
SA1
AR2
AR3
AR4
IC64_3D
IC58_9D
IC58_8D
IC9_6D
IC9_10D
IC9_13D
IC10_3D
IC10_6D
C76_4D
IC19_14
VIC76D 74LS04
VIC1 74LS273
VIC2 74LS273
IC76_8D
V1*
V2*
V4*
V8*
V16*
V32*
V64*
V128*
BCG[0..9]
VIC82 74LS273
VIC81 74LS273
VIC84 74LS273
VIC83 74LS273
IC82_11
IC84_11
CP1
IC83_6
IC83_9
BPL[0..3]
VIC23 74LS245
buffers
VIC3C 74LS86
VIC3D 74LS86
VIC37B 74LS86
VIC37C 74LS86
VIC67E 74LS04
IC37_8
IC67_10
VIC37A 74LS86
IC44_1D
V8
VIC4D 74LS86
VIC3A 74LS86
VIC3B 74LS86
VIC37B 74LS86
VIC60C 74LS08
VIC67D 74LS04
CP2
IC68_2D
IC10_10D
VIC80 74LS157
IC80_4
IC80_7
IC84_1D
VIC91C 74LS86
VIC91D 74LS86
JOTEGO
Sheet: /bottom/scroll/scroll RAM/
File: scram.kicad_sch
Title: Scroll RAM
Size: A3
Date:
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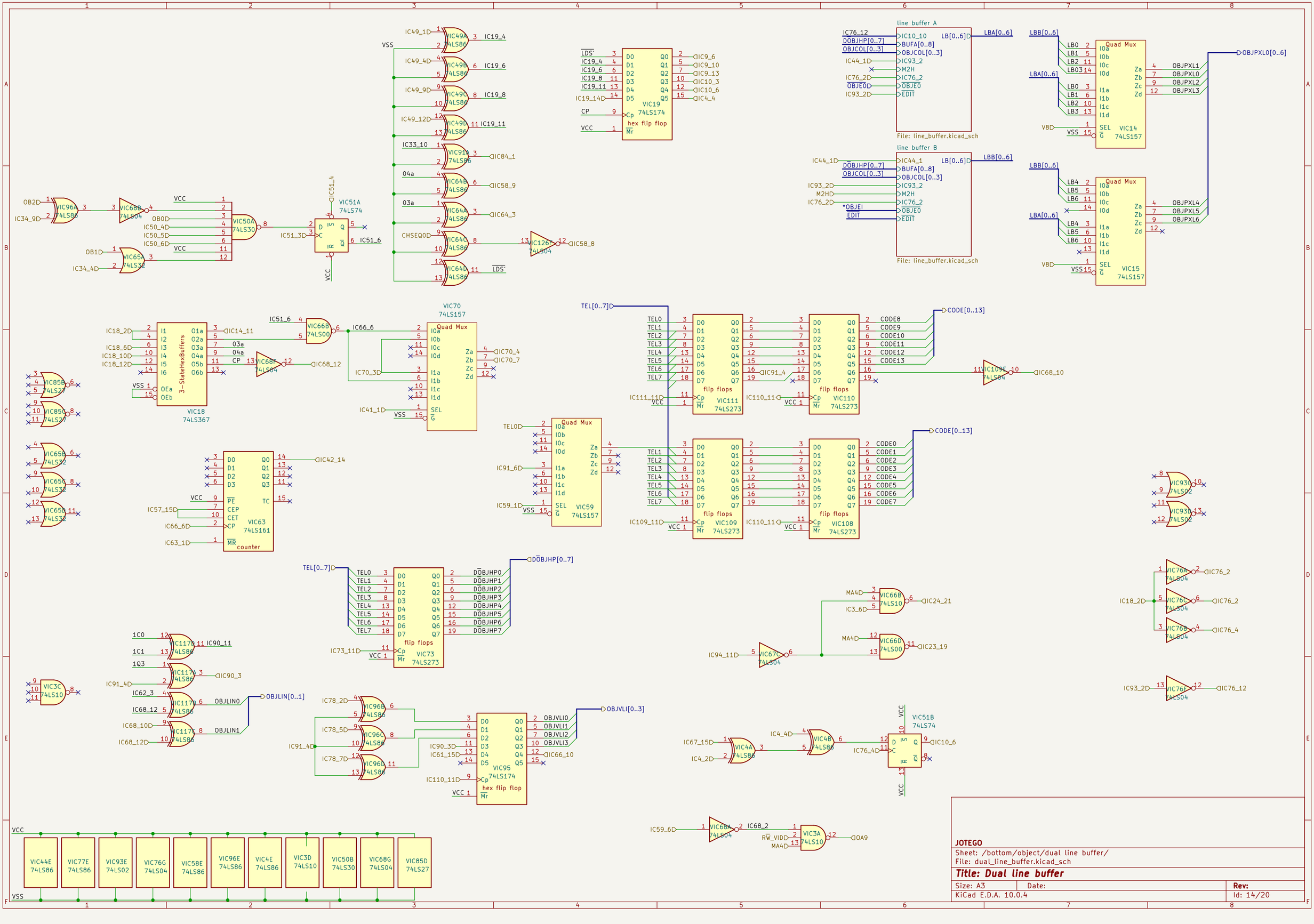


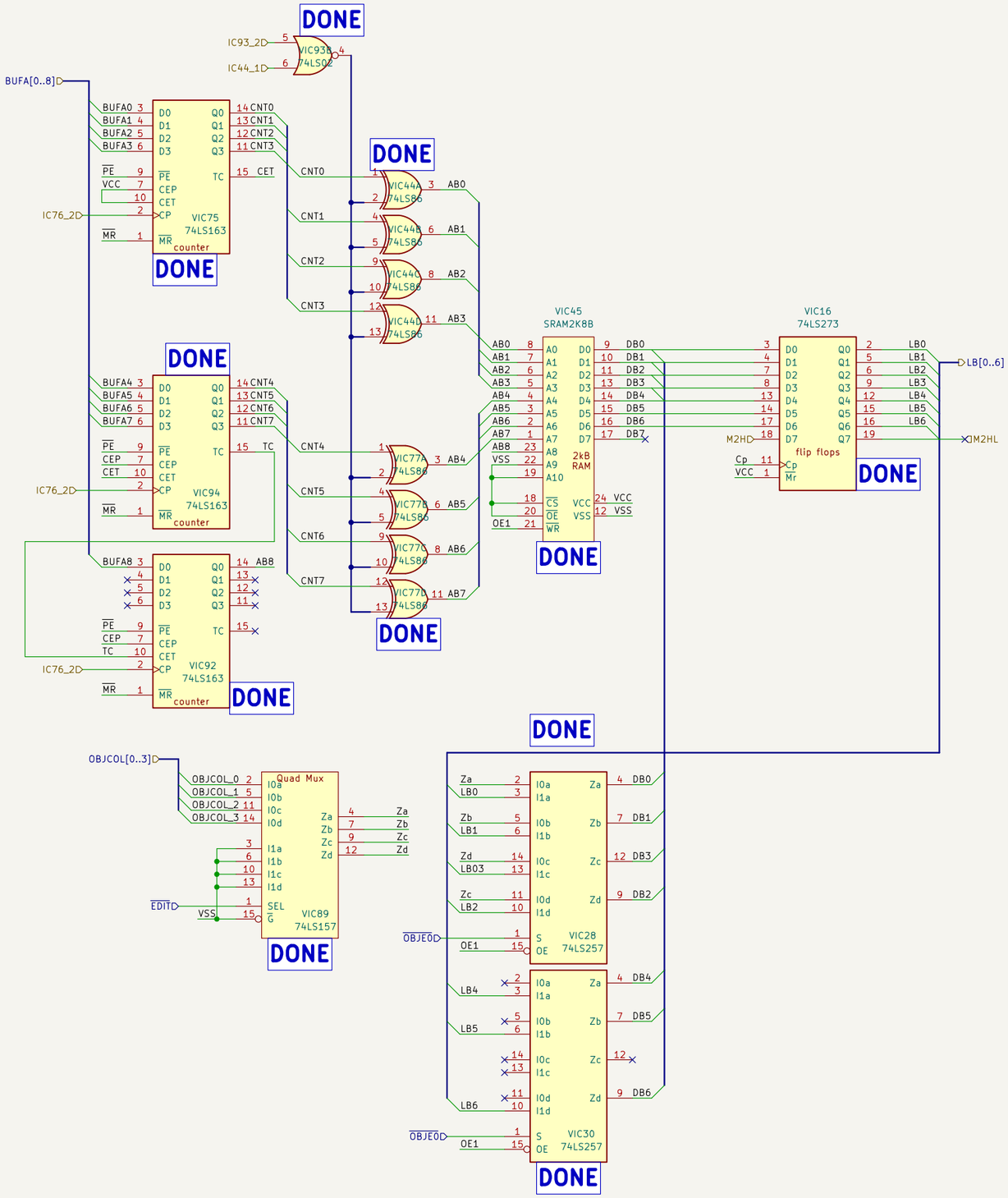
JOTEGO

Sheet: /bottom/object/
File: object.kicad_sch

Title: Objects

Size: A3	Date:	**Rev:**
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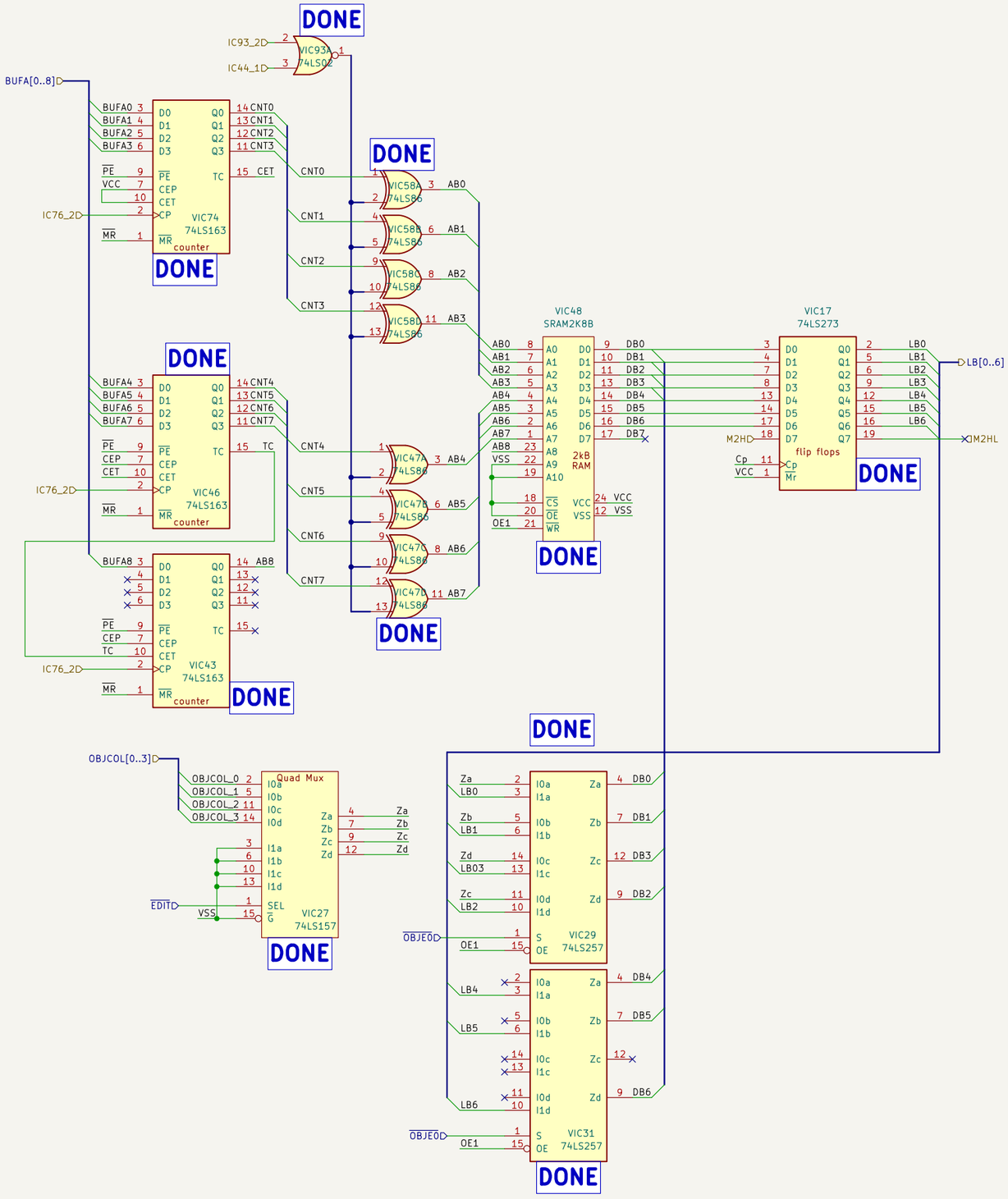

line buffer A
File: line_buffer.kicad_sch
line buffer B
File: line_buffer.kicad_sch
Quad Mux
flip flops
hex flip flop
3-StateHexBuffers
counter
JOTEGO
Sheet: /bottom/object/dual line buffer/
File: dual_line_buffer.kicad_sch
Title: Dual line buffer
Size: A3
Date:
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Sheet: /bottom/object/dual line buffer/line buffer A/
File: line_buffer.kicad_sch

Title: Line Buffer

Size: A3	Date:	Rev:
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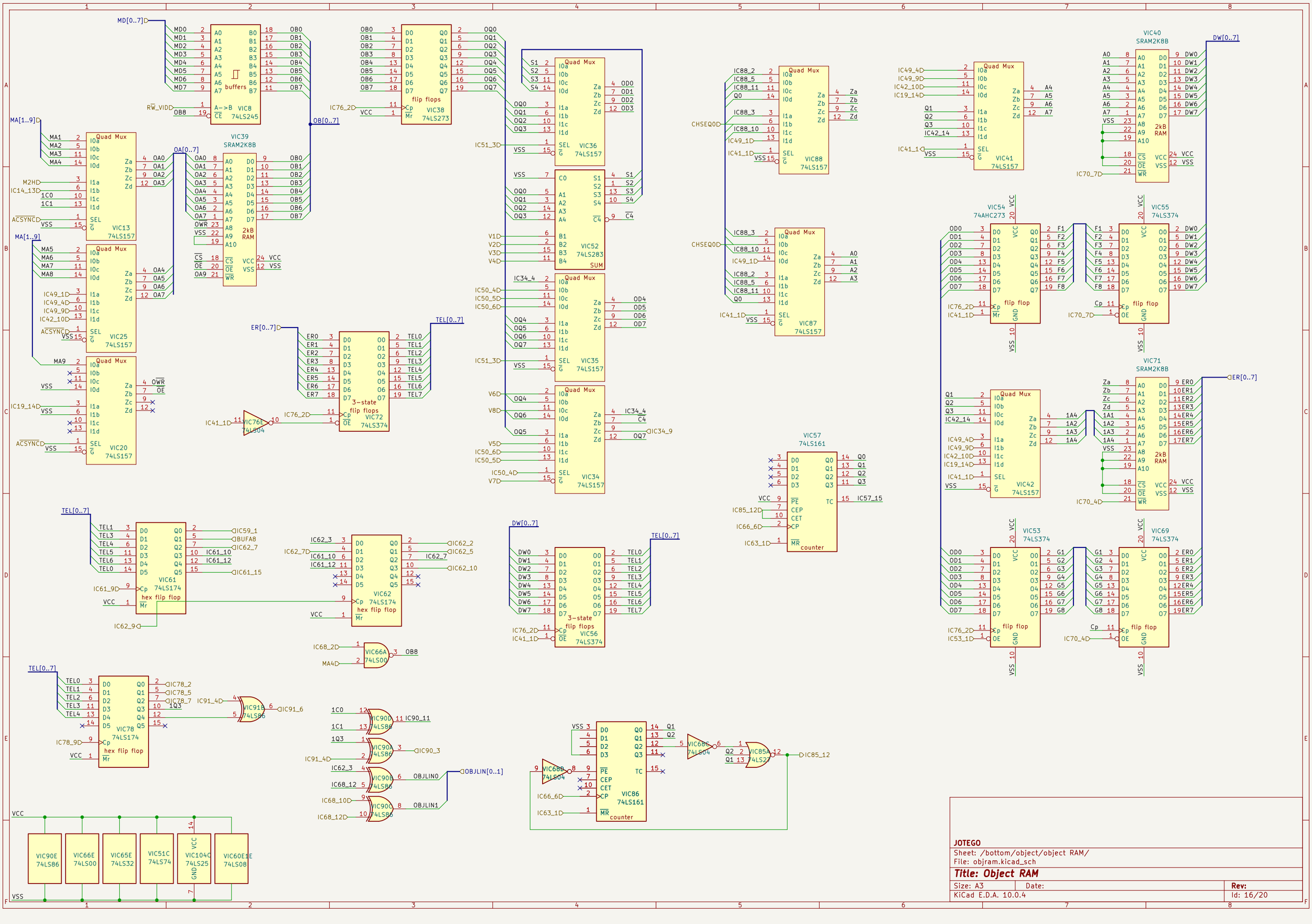
Sheet: /bottom/object/dual line buffer/line buffer B/
File: line_buffer.kicad_sch

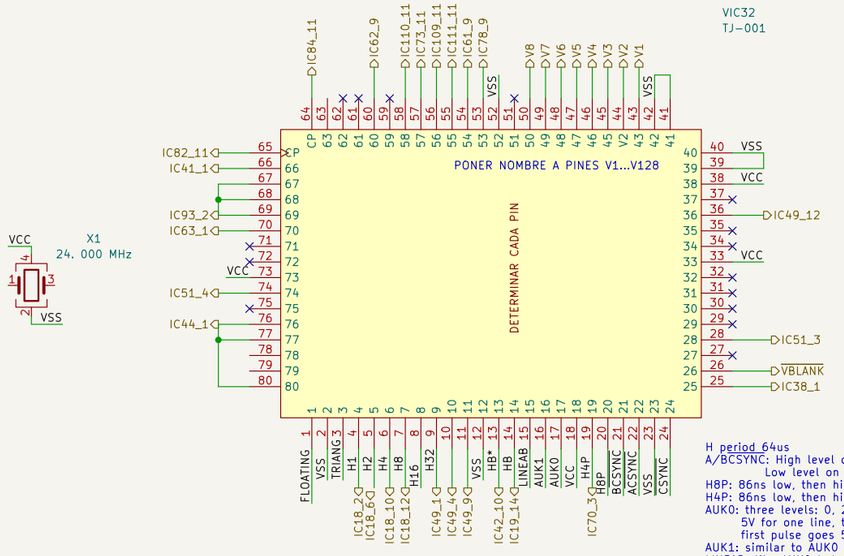
Title: Line Buffer

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VU117 TC534000AP
VIC114 TC534000AP
512kB MASK ROM
VIC115 24J
VIC116 24J
VIC118 24J
VIC119 24J
256 kB MASK ROM
VIC103 74LS194
VIC102 74LS194
VIC107 74LS194
VIC106 74LS194
shift register
VIC105 74LS157 Quad Mux
VIC104A 74LS25
VIC104B 74LS25
VIC60A 74LS08
VIC60B 74LS08
VIC60D 74LS08
VIC66C 74LS00
VIC64B 74LS04
VIC68A 74LS04
VIC101 74LS158
OBJVLI[0..3]
OBJLIN[0..1]
CODE[0..13]
OBJCOL[0..3]
OBJEO
*OBJEI
EDIT
V8
ACSYNCD
IC82_11D
IC66_10D
IC62_2D
HCLK1D
OROM_OED
CE
CODE_10
-> CE
2 -> A16
ambiar posiciones de pines y cables

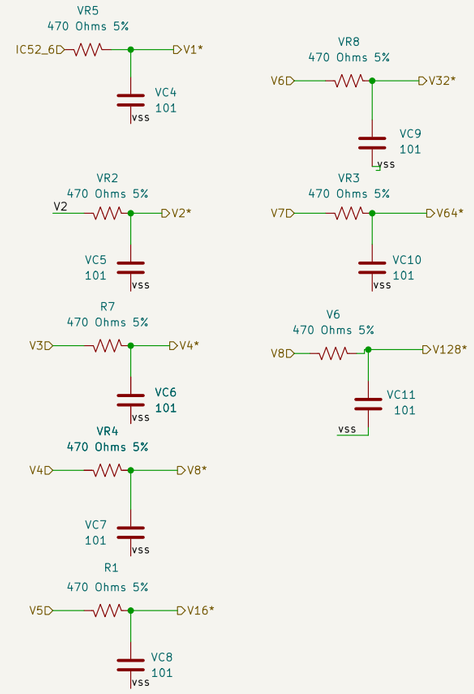
JOTEGO
Sheet: /bottom/object/object ROM/
File: objrom.kicad_sch
Title: Object ROM
Size: A3 | Date: | Rev:
KiCad E.D.A. 10.0.4 | Id: 15/20


MD[0..7]
buffers
VIC8
74LS245
RW_VIDD
OB[0..7]
flip flops
VIC38
74LS273
IC76_2
VIC39
SRAM2K8B
2kB
RAM
MA[1..9]
Quad Mux
VIC13
74LS157
ACSYNCD
M2H
IC14_13
VIC25
VIC20
OA[0..7]
OWR
OE
3-state
flip flops
VIC72
74LS374
ER[0..7]
TEL[0..7]
VIC76E
74LS04
IC41_1D
VIC36
VIC52
74LS283
SUM
VIC35
VIC34
IC51_3D
IC50_4D
IC50_5D
IC50_6D
IC34_4
IC34_9
VIC88
VIC87
CHSEQ0D
IC88_2
IC88_3
IC88_10
IC88_11
IC49_1D
IC41_1D
VIC41
VIC40
IC49_4D
IC49_9D
IC42_10D
IC19_14D
IC42_14
IC41_1D
IC70_7D
VIC54
74AHC273
VIC55
74LS374
flip flop
IC76_2D
IC41_1D
IC70_7D
VIC57
74LS161
counter
VIC42
VIC71
IC70_4D
VIC53
VIC69
IC53_1D
VIC61
74LS174
hex flip flop
IC61_9D
IC62_9D
IC59_1
BUFA8
IC62_7
IC61_10
IC61_12
IC61_15
VIC62
IC62_3
IC62_7D
IC61_10
IC61_12
IC62_2
IC62_5
IC62_10
VIC56
IC68_2D
MA4D
VIC66A
74LS00
OBB
VIC78
IC78_9D
IC78_2
IC78_5
IC78_7
1Q3
VIC91B
74LS86
IC91_4D
IC91_6
1C0
1C1
VIC90D
VIC90A
VIC90B
VIC90C
IC90_11
IC90_3
OBJLIN0
OBJLIN1
OBJLIN[0..1]
VIC86
IC66_6D
IC63_1D
VIC68B
VIC68A
VIC85A
IC85_12
VCC
VSS
VIC90E
VIC66E
VIC65E
VIC51C
VIC104C
VIC60E1E
JOTEGO
Sheet: /bottom/object/object RAM/
File: objram.kicad_sch
Title: Object RAM
Size: A3
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 16/20



TRIANG: triangular waveform at 12MHz
H1: 6MHz
H2: 3MHz
H4: 1.5MHz
H8: 750Hz
H16: 375Hz
H32: 187Hz
H64: 93Hz
H128: 47Hz

H period 64us
A/BCSYNC: High level on visible area 5V, during VBLANK ~2V
Low level on visibile area ~2V, during VBLANK 0V
H8P: 86ns low, then high. Period 668ns
H4P: 86ns low, then high. Period 332ns
AUK0: three levels: 0, 2 and 5V. subperiod 2.66us. period 128us
5V for one line, then pulses down during the next line. The first pulse goes 5->0V, the rest 5->2V
AUK1: similar to AUK0
LINEAB: like AUK0 but purely digital (0/5V)
HB: 64us period, 21.4us high



JOTEGO		
Sheet: /bottom/timing/		
File: timing.kicad_sch		
Title:		
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 13/20